

Zeyu Gan 甘泽宇

Ph.D. Candidate in Artificial Intelligence

Gaoling School of Artificial Intelligence, Renmin University of China

✉ zygan@ruc.edu.cn | github.com/ZyGan1999 | [Google Scholar](https://scholar.google.com/citations?user=zygan1999)

Beijing, China | Born June 1999



RESEARCH INTERESTS

Mechanism analysis and theory of Large Language Models (LLMs) — including chain-of-thought reasoning, slow-thinking and policy optimization, synthetic data, LLM alignment, and the theoretical foundations of machine learning.

EDUCATION

Ph.D. in Artificial Intelligence

Sep 2024 – Jun 2028 (expected)

Gaoling School of Artificial Intelligence, Renmin University of China · Advisor: Prof. Yong Liu

M.S. in Computer Application Technology

Sep 2021 – Jun 2024

School of Information, Renmin University of China · Advisor: Prof. Suyun Zhao

B.S. in Computer Science

Sep 2017 – Jun 2021

School of Information, Renmin University of China

PUBLICATIONS

- Zeyu Gan**, Huayi Tang, Yong Liu. [Statistical Priors for Implicit Preferences: Decoupling Skill Selection as a Local Harness in Personal Agents](#). *arXiv:2606.05828*, 2026.
- Zeyu Gan**, Ruifeng Ren, Wei Yao, Xiaolin Hu, Gengze Xu, Chen Qian, Huayi Tang, Zixuan Gong, Xinhao Yao, Pengwei Tang, Zhenxing Dou, Yong Liu. [Beyond the Black Box: A Survey on the Theory and Mechanism of Large Language Models](#). *arXiv:2601.02907*, 2026.
- Guoqing Wang, Sunhao Dai, Guangze Ye, **Zeyu Gan**, Wei Yao, Yong Deng, Xiaofeng Wu, Zhenzhe Ying. [Information Gain-based Policy Optimization: A Simple and Effective Approach for Multi-Turn LLM Agents](#). *arXiv:2510.14967*, 2025.
- Zeyu Gan**, Hao Yi, Yong Liu. [Why Does Reasoning Length Converge? Unveiling the Underfitting-Overfitting Trade-off in Chain-of-Thought](#). *arXiv:2509.04027*, 2025.
- Zeyu Gan**, Yun Liao, Yong Liu. [Rethinking External Slow-Thinking: From Snowball Errors to Probability of Correct Reasoning](#). *International Conference on Machine Learning (ICML)*, 2025. [ICML 2025](#)
- Zeyu Gan**, Yong Liu. [Towards a Theoretical Understanding of Synthetic Data in LLM Post-Training: A Reverse-Bottleneck Perspective](#). *International Conference on Learning Representations (ICLR)*, 2025. [ICLR 2025](#)
- Pan Du, Suyun Zhao, Puhui Tan, Zisen Sheng, **Zeyu Gan**, Hong Chen, Cuiping Li. [Towards a Theoretical Understanding of Semi-Supervised Learning Under Class Distribution Mismatch](#). *IEEE Transactions on Pattern Analysis and Machine Intelligence (TPAMI)*, 2025. [TPAMI](#)
- Zeyu Gan**, Suyun Zhao, Jinlong Kang, Liyuan Shang, Hong Chen, Cuiping Li. [Superclass Learning with Representation Enhancement](#). *IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2023. [CVPR 2023](#)

HONORS & AWARDS

- Outstanding Master's Thesis (top 10%), Renmin University of China — 2024
- Outstanding Graduate (top 10%), Renmin University of China — 2024
- Outstanding Postgraduate Scholarship (First Class, top 10%) — 2023
- College Student Innovation Experiment Program (National Project) — 2019

EXPERIENCE

Research Assistant, Gaoling School of AI

Sep 2024 – Present

Renmin University of China. Research on the mechanism and theory of LLMs, reasoning, and alignment.

Research Assistant, DW&BI Machine Learning Group

Sep 2021 – Jun 2024

School of Information, Renmin University of China. Machine learning, computer vision, and weakly-supervised learning.

Research Intern, Machine Learning Platform Dept. (TEG)

Aug 2023 – Nov 2023

Tencent. Worked on advertising recommendation algorithms; studied large-batch deployment and optimizer design.

ACADEMIC SERVICE

Conference Reviewer

[ICLR](#)

[ICML](#)

[CVPR](#)